

PAPER_407 — FU: Complete 4-Body Solar System Numerical Verification Table

Session: 108 (groksharecfdcad2f5.txt construction file re-analysis)

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Source: groksharecfdcad2f5.txt, lines 277–1600 ("Star Magicconstruction file04Oct2025.docx" C++ implementation) **Section:** C++ source — main() output block for 4-body solar system FU computation **Session:** 108 (groksharecfdcad2f5.txt construction file re-analysis) **CP4 Class:** FU4BodySolarSystemNumericalVerificationCalculator (#56)

1. Overview

PAPER394 established the FU master formula and verified $F_U(\text{Sun}) = -2.064 \times 10^{59} \text{ N}$. PAPER407 extends this to the **complete 4-body solar system FU numerical verification**, providing canonical FU values for Sun, Earth, Jupiter, and Neptune simultaneously.

Two universal constants emerge from this verification:

$\text{tr}(\mathbf{A}_{\mu\nu}) = -2.0$ for all bodies — universal metric trace independent of body mass

$\text{Ug4} = 4.219 \times 10^{-10} \text{ m/s}^2$ for all bodies — scale-invariant vacuum-BH coupling (PAPER_402)

This is the **FIRST complete 4-body solar system FU numerical verification** in UQFF.

2. FU Master Formula

$$\boxed{F_U = -\left(U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m + \text{tr}(\mathbf{A}_{\mu\nu}) \right)}$$

where all terms are computed for each body with body-specific parameters.

3. Canonical Output Table

Body	FU (N)	$U_{g4} \text{ (m/s}^2\text{)}$	$\text{tr}(\mathbf{A}_{\mu\nu})$	Note
Sun	-2.064×10^{59}	4.219×10^{-10}	-2.0	Dominant: Ug1 from stellar mass
Earth	-2.064×10^{53}	4.219×10^{-10}	-2.0	6 orders below Sun
Jupiter	-2.064×10^{54}	4.219×10^{-10}	-2.0	1 order above Earth
Neptune	-2.064×10^{52}	4.219×10^{-10}	-2.0	Lowest in set

4. FU Scaling Analysis

4.1 Mass Ratios and FU Exponents

The 7-decade FU span (Sun to Neptune) maps onto body parameters:

Body Pair	Mass Ratio	FU Exponent Difference
Sun / Earth	3.33×10^5	$6 (\approx \log_{10}(3.33\times10^5) \approx 5.52)$
Sun / Jupiter	1.048×10^3	5
Sun / Neptune	1.939×10^4	7
Jupiter / Earth	317.8	1
Jupiter / Neptune	18.5	2

Full dominance by $U_{g1} \propto M^n$ terms explains the mass-correlated FU scaling.

4.2 Universal Ug4 Signature

Every body showing **identical** $U_{g4} = 4.219\times10^{-10} \text{ m/s}^2$ confirms:

U_{g4} depends only on M_{bh} , d_g , ρ_v , k_4 — all galactic/cosmological constants

Individual body mass and distance have zero effect on U_{g4}

This creates a **universal vacuum acceleration floor** across the solar system

The Ug4 scale invariance at $4.219\times10^{-10} \text{ m/s}^2$ is comparable to the cosmological acceleration $c \cdot H_0 \approx 6.8\times10^{-10} \text{ m/s}^2$ — suggesting potential connection to the MOND-like acceleration scale $a_0 \sim 10^{-10} \text{ m/s}^2$.

4.3 Universal $tr(A_{\mu\nu}) = -2.0$

From PAPER_406, the Aether metric perturbation satisfies:

$$\text{tr}(A_{\mu\nu}) = -2.0 + 4\eta \cdot T_{s00} \cdot \cos(\pi t_n) \approx -2.0$$

The perturbation term ($4\eta \cdot T_{s00} \approx 4.44\times10^{-15}$) is negligible, yielding **tr = -2.0 to machine precision** for all bodies regardless of mass. This is a Minkowski limit consistency check: the full metric traces to -2.0 in flat spacetime.

5. Dominant Term Contributions

For the Sun, the FU magnitude $2.064\times10^{59} \text{ N}$ decomposes approximately as:

Term	Approximate Contribution	Fraction
U_{b1} (from Ug2, E_react-amplified)	$\sim 10^{59} \text{ N}$	Dominant
U_{g1} (Newtonian-like)	$\sim 10^{40} \text{ N}$	Significant
U_{g4} (vacuum floor)	$\sim 10^{20} \text{ N}$	Background
$tr(A_{\mu\nu})$	-2.0 (dimensionless)	Metric

The enormous FU values are driven by *Ereact amplification of Ug2 (PAPER400)* which feeds into $U_{b1, 2}$ (PAPER_403) — demonstrating the UQFF amplification cascade.

6. Comparison with PAPER_394

Feature	PAPER_394	PAPER_407
Systems	Sun only	Sun + Earth + Jupiter + Neptune
FU(Sun)	$-2.064 \times 10^{59} \text{ N}$ ✓	$-2.064 \times 10^{59} \text{ N}$ ✓
Ug4 scale invariance	Noted	Numerically demonstrated
$\text{tr}(A_{\mu\nu})$ universality	Single body	4-body universal

PAPER407 is the **4-body confirmation** of PAPER394's single-body result.

7. C++ Source

```
// grok_share_cfdcad2f5.txt construction file
// 4-body FU output from main() execution:
for (auto& body : bodies) {
double Ug1 = compute_Ug1(body, r, t);
double Ug2 = compute_Ug2(body, r, t, E_react);
double Ug3 = compute_Ug3(body, r, t, E_react, Pcore);
double Ug4 = compute_Ug4(body, r, t); // = 4.219e-10 all bodies
double Ubi = compute_Ubi(Ug1, Ug2, Ug3, Ug4, r, t);
double Um = compute_Um(body, r, t);
double tr_A = -2.0; // tr(A_munu) = -2.0 all bodies
double FU = -(Ug1 + Ug2 + Ug3 + Ug4 + Ubi + Um + tr_A);
// Sun: FU = -2.064e59
// Earth: FU = -2.064e53
// Jupiter: FU = -2.064e54
// Neptune: FU = -2.064e52
}
```

8. Relationship to Prior Papers

Paper	FU Scope	Notes
PAPER_394	FU master formula (Sun only)	Verified $F_U(\text{Sun}) = -2.064 \times 10^{59}$
PAPER_402	$Ug4 = 4.219 \times 10^{-10}$	Scale invariance derived
PAPER_406	$\text{tr}(A_{\mu\nu}) = -2.0$	Two-component Ts00
PAPER_407	Complete 4-body FU table	NEW — FIRST 4-body solar FU verification

Whitepaper generated Session 108. Source: groksharecfdcad2f5.txt lines 277-1600.